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# **Machine Learning** (911.236), **Statistical Learning Theory** (536.505), **Advanced Machine Learning** (911.936)

Exercise sheet **A**

**Exercise 1.** 3 P.

Suppose the response time $X$ (in seconds) of a database query is a non-negative random variable with the following probability mass function (PMF):

| **Response time $x$ (in [s])** | 0 | 1 | 3 | 5 | 10 |
|---|---|---|---|---|---|
| **Probability** $\mathbb{P}(X = x)$ | 0.15 | 0.35 | 0.25 | 0.15 | 0.10 |

1. Compute the expected value $\mathbb{E}[X]$.
2. Use Markov's inequality to upper bound $\mathbb{P}(X \geq 6)$.
3. Compute the exact value of $\mathbb{P}(X \geq 6)$ using the PMF.
4. Compare the two values and briefly comment on the tightness of the Markov bound in this example.

**Exercise 2.** 3 P.

Consider a learning problem with $n$ binary input variables $v_1, \ldots, v_n$ with $v_i \in \{0, 1\}$. Hence the instance space is

$$\mathcal{X} = \{0, 1\}^n$$

and the label space is $\mathcal{Y} = \{0, 1\}$.
Assume the hypothesis class $\mathcal{H}$ contains *all possible* Boolean functions from $\mathcal{X}$ to $\mathcal{Y}$.

1. How many different functions exist in $\mathcal{H}$?
2. Using the PAC learnability result for finite hypothesis classes, recall that the sample complexity satisfies

$$m \geq \frac{1}{\epsilon}\left(\log|\mathcal{H}| + \log\frac{1}{\delta}\right).$$

   Insert the size of $\mathcal{H}$ into this expression.
3. What happens to the sample complexity as $n$ increases?
4. Briefly explain why this hypothesis class is problematic from a statistical learning perspective.

**Total #points: 6 P.**